

UNIVERSITY OF BOTSWANA
DEPARTMENT OF MATHEMATICS
2015/2016 SEMESTER II
EXAMINATIONS

COURSE NUMBER: MAT 111 DURATION: 2 HRS. DATE: May 2016

TITLE OF PAPER: INTRODUCTORY MATHEMATICS I

SUBJECT: MATHEMATICS TITLE OF EXAMINATION: BSc. I

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO QUESTIONS IN SECTION B.
- ALL MARKS ARE INDICATED IN BRACKETS [].

NO. OF PAGES

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SECTION A

Answer all Questions
Each question carries 10 marks.

Question 1

(a) Find $\cos\left(\frac{7\pi}{6}\right)$. [4]

(b) If $\sin x = -\frac{2}{5}$, find the value of $\cot x$. [6]

Question 2

(a) Simplify

$$\frac{(\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3}) + \sqrt{2}}{\sqrt{2} - 1}.$$
 [3]

(b) Simplify

$$\frac{(3t^2)^{-3}t^3}{(4t)^{-2}}.$$
 [4]

(c) Factorize $27 - 8y^3$. [3]

Question 3

(a) Determine the domain of [6]

$$f(x) = \frac{1}{\sqrt{36 - x^2}}.$$

(b) Find a number k such that $f \circ g = g \circ f$ given that $f(x) = \frac{kx + 1}{2}$ and $g(x) = 2x - 3$. [4]

Question 4

Solve the following

(a) $\frac{2}{x+1} - \frac{1}{x^2-1} = 0$ [4]

(b) $|2x - 1| - x = 0$ [6]

Question 5

- (a) Rationalize the denominator of

$$\frac{2 + \sqrt{6}}{\sqrt{6} - 1}.$$

[3]

- (b) Determine whether the following function is odd, even or neither:

$$f(x) = -3x^2 - 2$$

[3]

- (c) Find the remainder when the polynomial

$$x^5 - 2x^3 - 5x + 7$$

is divided by $x^2 + 1$.

[4]

Question 6

- (a) Find the domain and the range of $g(x) = 5^x + 1$.

[5]

- (b) Solve

$$\log x + \frac{1}{\log_5 10} = \log(2x + 1).$$

[5]

SECTION B

Answer TWO (2) Questions out of FOUR (4)
Each Question carries 20 marks.

Question 7

(a) Solve $9^x - 6(3^x) + 8 = 0$. [7]

(b) Solve $\frac{x^2 + x - 6}{x - 1} \leq 0$. [8]

(c) Find the exact value of $\cos\left(\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)\right)$. [3]

(d) Express $\sin 3x + \sin 2x$ as a product. [2]

Question 8

(a) For $0 \leq x \leq \pi$, find all solutions of

$$\sin^2 x - \cos^2 x = \frac{1}{2}$$

[10]

(b) If $x - 1$ and $x - 2$ are factors of $P(x) = x^3 + ax^2 + bx + 2$, find the constants a and b . [6]

(c) Simplify $\log_2 7 \times \log_7 9 \times \log_3 2$. [4]

Question 9

(a) Given that the function $f(x) = -2x^2 + 5x - 3$. Find

(i) Find the y -intercept and the x -intercepts of the function $f(x)$. [4]

(ii) Find the vertex. [4]

(iii) Sketch the function $y = f(x)$. [5]

(b) Find the equation of the line perpendicular to $y + 2x = 4$ passing through $(-1, 2)$. [4]

(c) Prove that

$$\cos 2\theta = 2 \cos^2 \theta - 1.$$

[3]

Question 10

(a) Given that

$$f(x) = \frac{3x + 1}{5x - 11},$$

- (i) show that f is an injective function.
- (ii) find the inverse function f^{-1} .
- (iii) what is the domain of f^{-1} ?

(b) Find the asymptotes of the function

$$f(x) = \frac{x + 1}{x - 2}.$$

(c) Prove that

$$\frac{\sin x}{1 + \cos x} = \csc x - \cot x.$$